Code 2523

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NREIP
NAVAL RESEARCH ENTERPRISE INTERNSHIP PROGRAM

Project Objective and Intern Contribution:
The project's aim is:
To work towards the creation of an AI/ML based search engine to be used by sailors to find relevant troubleshooting information while on deployment.

The methods we used to accomplish this aim were:
Using purpose-built AI/ML models to compile and organize technical information for easy searching.

I was assigned to:
The SWFTS Architecture team and SWFTS Integration and Test team

My contributions were….
- **Using AI models and other open source software to develop a Python-based application meant to extract and transcribe audio and visual data from relevant training videos into text data.**
- **I containerized my Python-based application, making it portable and reliable to use so it may be used in the future for similar use cases.**
- **I generated training data for the AI/ML based search engine.**
- **Aiding the Integration and Testing team during their operations testing procedures, specifically with the navigation system.**

Answer: I am most proud knowing that my contributions will be used after my internship is over.

2. Question: Why was the internship valuable?

Answer: I found this internship valuable since I was constantly working at a challenging level that forced me to learn new skills. At the same time, my work was necessary for my team so, even though my more experienced coworkers could have gotten it done faster and with fewer mistakes, my contributions will make an actual difference in the future. It also allowed my coworkers to focus their attention on higher priority projects.

3. Question: Advice for future cohorts?

Answer: Always ask for help and never assume there is nothing for you to do. Have fun and listen. Try different things and learn what you like most.

Results / Accomplishments / Next Steps:
We demonstrated:
AI and machine learning can be used successfully and reliably to compile information for uses relevant to the fleet.
The impact for the Navy is:
Sailors on deployment will have easier access to information regarding the use and safety of SWFTS and all related subsystems on their vessel.
What's most important is:
Finding ways to improve on previous developments to continually innovate how our work is done.
In the future this work will be able to:
Develop similar systems to further aid sailors on deployment and develop more advanced technical information systems.